

First Order Linear Differential Equations

Worksheet 3

Today's Goals

- Recognize a first order **linear** equation $\frac{dy}{dx} + P(x)y = Q(x)$.
- Compute the integrating factor $\rho(x) = e^{\int P(x) dx}$.
- Use $\frac{d}{dx}[\rho(x)y] = \rho(x)Q(x)$ to integrate both sides.
- Solve general and initial-value problems, including ones requiring substitution to integrate.

We solve linear first order equations of the form

$$\frac{dy}{dx} + P(x)y = Q(x),$$

where $P(x)$ and $Q(x)$ are continuous. We cannot separate variables, but multiplying by the **integrating factor** $\rho(x) = e^{\int P(x) dx}$ makes the left side an exact derivative.

Teacher note. Since $\frac{d}{dx}(\int P(x) dx) = P(x)$, the product rule gives

$$\frac{d}{dx} \left[y e^{\int P(x) dx} \right] = e^{\int P(x) dx} \frac{dy}{dx} + P(x) e^{\int P(x) dx} y = Q(x) e^{\int P(x) dx}.$$

Integrating, $y e^{\int P(x) dx} = \int Q(x) e^{\int P(x) dx} dx + C$.

Steps to solve $\frac{dy}{dx} + P(x)y = Q(x)$:

1. Calculate $\rho(x) = e^{\int P(x) dx}$.
2. Multiply the equation by $\rho(x)$.
3. Notice that $\frac{d}{dx}[\rho(x)y] = \rho(x)Q(x)$.
4. Integrate both sides: $\rho(x)y = \int \rho(x)Q(x) dx + C$.
5. Solve: $y = \frac{1}{\rho(x)} \left[\int \rho(x)Q(x) dx + C \right]$.

Using an Integrating Factor

1. Solve $2xy' + y = 10\sqrt{x}$ for $x > 0$.

Solution. Put the equation in standard form by dividing by $2x$:

$$y' + \frac{1}{2x} y = \frac{5}{\sqrt{x}}, \quad P(x) = \frac{1}{2x}, \quad Q(x) = \frac{5}{\sqrt{x}}.$$

The integrating factor is

$$\rho(x) = e^{\int \frac{1}{2x} dx} = e^{\frac{1}{2} \ln x} = (e^{\ln x})^{1/2} = x^{1/2}.$$

Multiplying by $\sqrt{x}$:

$$\sqrt{x} y' + \frac{1}{2\sqrt{x}} y = 5, \quad \text{i.e.} \quad \frac{d}{dx}(\sqrt{x} y) = 5.$$

Integrating, $\sqrt{x} y = 5x + C$, so

$$y = \frac{1}{\sqrt{x}}(5x + C) = 5\sqrt{x} + \frac{C}{\sqrt{x}} \quad (\text{general solution}).$$

2. Solve the initial value problem $y' = 2xy + 3x^2e^{x^2}$, $y(0) = 5$.

Solution. In standard form, $y' - 2xy = 3x^2e^{x^2}$, so $P(x) = -2x$ and $Q(x) = 3x^2e^{x^2}$.

$$\rho(x) = e^{\int -2x dx} = e^{-x^2}.$$

Multiplying through:

$$e^{-x^2} y' - 2xe^{-x^2} y = 3x^2e^{x^2} \cdot e^{-x^2} = 3x^2, \quad \frac{d}{dx}(e^{-x^2} y) = 3x^2.$$

Integrating, $e^{-x^2} y = x^3 + C$, so $y = e^{x^2}(x^3 + C)$ (general solution). Applying $y(0) = 5$:

$$5 = e^0(0 + C) = C,$$

so $y = e^{x^2}(x^3 + 5)$ (particular solution).

3. Solve $y' = -y \tan x + (\cos^2 x) \sin x$, $y(0) = 3$, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

Solution. In standard form, $y' + (\tan x)y = (\cos^2 x) \sin x$, so $P(x) = \tan x$ and $Q(x) = (\cos^2 x) \sin x$. With $u = \cos x$, $du = -\sin x dx$:

$$\rho(x) = e^{\int \tan x dx} = e^{\int \frac{\sin x}{\cos x} dx} = e^{-\int \frac{du}{u}} = e^{-\ln u} = \frac{1}{\cos x} = \sec x,$$

valid since $\cos x > 0$ on $-\frac{\pi}{2} < x < \frac{\pi}{2}$. Multiplying through:

$$(\sec x)y' + (\sec x)(\tan x)y = (\sec x)(\cos^2 x) \sin x = (\cos x) \sin x,$$

i.e. $\frac{d}{dx}[(\sec x)y] = (\cos x) \sin x$. Integrating with $u = \cos x$:

$$(\sec x)y = \int (\cos x) \sin x dx = -\int u du = -\frac{u^2}{2} + C = -\frac{\cos^2 x}{2} + C.$$

Thus $y = -\frac{\cos^3 x}{2} + C \cos x$ (general solution). Applying $y(0) = 3$:

$$3 = -\frac{\cos^2 0}{2} + C \cos 0 = -\frac{1}{2} + C \implies C = \frac{7}{2},$$

so $y = -\frac{\cos^3 x}{2} + \frac{7}{2} \cos x$ (particular solution).